

Kathmandu University

Assignment - II

Group – M. Sc.

Subject: MATH 509

Text Book: Mathematical Analysis
(2nd Edition)

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A

Define the following:

Metric space, Complete metric space, Convergent Sequences in a Metric Space, Cauchy Sequence, Complete metric space, continuous and uniformly continuous function in metric space,

Prove that the following theorems:

- a. Theorem 3.34
- b. Theorem 3.38
- c. Theorem 4.2
- d. Theorem 4.8
- e. Theorem 4.10
- f. Theorem 4.17
- g. Theorem 4.22
- h. Theorem 4.23

B

Define the following:

Partition, Functions of Bounded Variation, total variation

Prove that the following theorems

- a. Theorem 6.2
- b. Theorem 6.6
- c. Theorem 6.9
- d. Theorem 6.11
- e. Theorem 6.12

C

Define the following:

Riemann-Stieltjes sum of f with respect to α , Step function, Sequences of Functions, point wise convergent, uniform convergent sequence of function and uniform convergent series of function

Prove that the following theorems

- a. Theorem 7.2
- b. Theorem 7.3
- c. Theorem 9.2
- d. Theorem 9.3
- e. Theorem 9.5

D

Define the measurable sets, Sets of measure zero, Borel sets and measurable function.